Problem-Solution Fit

Date	29 September 2026
Team ID	SWTID-2026-8642
Project Name	Import Data Using Transform Maps (Spreadsheet)
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience.

1. CUSTOMER SEGMENT(S) [CS]	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL]	5. AVAILABLE SOLUTIONS PROS & CONS [AS]
• System Administrators & ServiceNow/Database Developers. • Data Analysts responsible for migrating bulk business records. • Business Operations Teams uploading daily/weekly spreadsheet reports.	• Technical skill gap among non-developer users regarding complex script mappings. • File size constraints and web browser session timeout limits during large uploads. • Restricted system admin access permissions for executing direct database scripts.	• **Manual SQL/Database Scripts:** Pros: High flexibility. Cons: Extremely slow, requires developer skills, high failure risk. • **Basic CSV Import Tools:** Pros: Simple to upload. Cons: No transform mapping, no duplicate checking (coalesce), limited error logging.

2. PROBLEMS / PAINS + ITS FREQUENCY **[PR]**

• High error rates due to column header mismatch between Excel and target database (Daily).

• Duplicate records created due to improper coalesce/matching rules (Frequent).

• Time-consuming manual data cleaning and script writing for field mapping (Every Upload)

9. PROBLEM ROOT / CAUSE [RC]

• Lack of automated auto-mapping logic between source spreadsheet headers and target table schemas.

• Absence of automated pre-upload validation and robust error log tracking mechanisms

7. BEHAVIOR + ITS INTENSITY **[BE]**

• Users manually double-check every Excel column against target tables (High Effort).

• Repeatedly running test imports on staging tables to spot duplicate records (High Frequency).

3. TRIGGERS TO ACT [TR]

• Need to bulk import thousands of customer, asset, or transaction records quickly.

• High rate of system failures or corrupt data after raw spreadsheet uploads.

• Need to automate repetitive manual mapping tasks to save operational time.

10. YOUR SOLUTION [SL]

• An automated **Transform Map System** that ingests spreadsheet files into staging tables, auto-suggests field mappings, executes coalesce logic to block duplicates, and logs row-level import errors clearly.

8. CHANNELS OF BEHAVIOR [CH]

ONLINE

Web Application Portals, Cloud Import Set Dashboards, Email Notifications.

OFFLINE

Microsoft Excel Desktop App, Offline CSV Reports,

4. EMOTIONS
BEFORE / AFTER **[EM]**

• **Before:** Frustrated, anxious about corrupting database records, exhausted by manual mapping.

• **After:** Confident, relieved, satisfied with automated auto-mapping and clean data execution.

Data Validation Spreadsheets